YUHE (MAGGIE) DENG
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EDUCATION

MIT SLOAN SCHOOL OF MANAGEMENT Cambridge, MA
Candidate for Master of Finance, February 2028 2026 - Present

- Intended concentration in Financial Engineering
- Anticipated Courses: Financial Engineering, Options and Futures Markets, Deep Learning, AI and Money
- Clubs: Quantitative Finance Club, Data Analytics Club

BRANDEIS UNIVERSITY Waltham, MA
B.S. in Applied Mathematics and B.S. in Quantitative Economics, summa cum laude 2022 - 2026

- GPA: 4.0/4.0, Dean's List All Semesters
- Relevant Coursework: Linear Algebra, Differential Equations, Probability and Statistics, Real Analysis, Stochastic Processes, Numerical Methods, Microeconomic and Macroeconomic Theory, Econometrics
- Honors: Arnold Shapiro Prize in Mathematics, Phi Beta Kappa Honors Society

EXPERIENCE

EMPIRICAL PANDEMIC UNEMPLOYMENT INSURANCE THESIS Waltham, MA
Independent Researcher, Departmental High Honors 2025 - 2026

- Estimated the causal labor-market effects of early pandemic unemployment-insurance termination using a contiguous-border county-pair difference-in-differences design; unemployment fell about 0.5 percentage points, but employment, wages and establishment count showed no robust gains
- Constructed monthly and quarterly county panels from BLS administrative database (LAUS, QCEW) and 2020 Census data, pairing 492 border counties into 502 cross-state pairs, with a 907-pair alternative sample for robustness check
- Estimated event-study models with county and pair-by-period fixed effects and state-clustered standard errors, testing pre-treatment trends and robustness to county-level clustering

LAKEFRONT ASSET MANAGEMENT Beijing, China
Commodity Research Intern Summer 2025

- Developed a bottom-up industrial silicon cost model across 4 regions and 5 leading producers, decomposing raw material, electricity, labor expenses, and regional production structures to estimate producer-level cost curves and margins
- Stress-tested producer margins under changes in electricity costs, raw-material prices and market prices, identifying manufacturers and regions most exposed to commodity-price shocks
- Built and backtested ARIMA and GARCH models on 2.6 years of industrial silicon futures data; ARIMA matched a random-walk benchmark (1.2% daily MAPE), while GARCH reduced 95% risk-band breaches by 23% versus constant volatility
- Presented a research report to the investment team, integrating cost-curve estimates, price forecasts, supply-demand conditions, regional production trends, and policy effects to support market assessment and investment discussions

SEGREGATION DYNAMICS COMPUTATIONAL MODELING RESEARCH Waltham, MA
Project Lead Spring 2025

- Modeled the emergence of residential segregation from individual location preferences using the Schelling framework, in which agents relocate when the share of similar neighbors falls below a specified tolerance threshold
- Built a Python agent-based simulation incorporating heterogeneous tolerance thresholds, local neighborhood interactions, and stochastic relocation, tracking convergence time and neighborhood sameness over 20,000-iteration runs
- Classified 4 distinct dynamic regimes by convergence behavior and tolerance asymmetry between groups, using heat maps and log-scaled convergence-time plots across tolerance combinations
- Led the project from design to presentation, running weekly check-ins to track progress and resolve issues early, keeping communication open across the team, and integrating each member's simulations and analysis into one final report

BANK OF COMMUNICATIONS Hong Kong
Global Banking Intern Summer 2024

- Collaborated across compliance teams to screen 10,000+ cross-border transactions for state-owned enterprise subsidiaries, flagging compliance exceptions during account onboarding under regulatory deadlines
- Partnered with credit teams to assess corporate credit risk, analyzing financial statements and updating 5+ leverage, liquidity, and repayment-capacity metrics across two lending reports

ADDITIONAL INFORMATION

- Technical: Python, R, Stata, MATLAB, LaTeX
- Interests: skiing, yoga, Peking opera, calligraphy